

# LegalEase AI

## Micro-Engineering Project Prompt Document

|              |                                                                     |
|--------------|---------------------------------------------------------------------|
| Version      | v1.0 — Initial Release                                              |
| Audience     | Product Managers, Full-Stack Engineers, AI/ML Engineers, Legal SMEs |
| Jurisdiction | Republic of India   BNS, BNSS, BSA, IT Act 2000                     |
| AI Model     | Large Language Model (LLM) — RAG-enhanced, domain-tuned             |
| Modules      | 5 Core Feature Modules + 2 Cross-Cutting Guardrail Modules          |
| Deliverable  | Production-ready AI Legal Information Assistant                     |

**Mission:** Bridge the gap between complex Indian legal frameworks and the general public by providing accurate, empathetic, and actionable legal information — fully grounded in the Bharatiya Nyaya Sanhita (BNS), BNSS, BSA, and the IT Act 2000.

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## ■■ Project Overview & Architecture

LegalEase AI is an empathetic, knowledge-intensive AI assistant built on a Retrieval-Augmented Generation (RAG) architecture. It ingests authoritative Indian legal corpora — BNS, BNSS, BSA, IT Act 2000, Consumer Protection Act 2019, POSH Act 2013, and Model Tenancy Act 2021 — and delivers plain-language legal awareness to Indian citizens across socioeconomic strata.

### High-Level Architecture

| Layer          | Component                       | Responsibility                                          |
|----------------|---------------------------------|---------------------------------------------------------|
| Frontend       | React / Flutter Web             | Chat UI, voice input, multilingual toggle               |
| API Gateway    | FastAPI / Node.js               | Session mgmt, rate limiting, auth                       |
| AI Core        | LLM + RAG Pipeline              | Intent detection, context retrieval, response synthesis |
| Knowledge Base | Vector DB (Pinecone / Weaviate) | Chunked legal corpus with metadata                      |
| Safety Layer   | Rule Engine + Classifiers       | Guardrails, disclaimer injection, emergency routing     |
| Integrations   | Govt APIs / SMS / Email         | Complaint portals, helpline triggers                    |

### Core Design Principles

- Accuracy-first: All responses grounded in cited legal sections — no hallucinations.
- Empathy-by-design: Tone calibrated to reduce user anxiety and increase clarity.
- Safety-gated: Every legal pathway response appended with a mandatory disclaimer.
- Scalability: Stateless API design; horizontal scaling for concurrent user load.
- Accessibility: Multilingual support (Hindi, Malayalam, Tamil, Bengali, etc.) via translation layer.

## ■ Module 1 — Legal Rights Awareness

### M Legal Rights Awareness Engine

#### ■ Scope

- Consumer Rights under the Consumer Protection Act, 2019
- Women's Statutory Rights (maternity, protection from harassment)
- Labour & Workplace Rights (Wages Code, Industrial Disputes Act)
- Property & Inheritance Rights (Transfer of Property Act, Hindu Succession Act)
- Senior Citizen Welfare (Maintenance & Welfare of Parents Act, 2007)

#### ■ Functional Requirements & Actions

**Step 1.** Intent Classification: NLP classifier detects the legal theme from free-text input (e.g., 'My employer hasn't paid me' → Labour Rights).

**Step 2.** Act Mapping: RAG pipeline retrieves relevant act, chapter, and section from the knowledge base (e.g., Code on Wages, 2019 — Section 17 for delayed payment).

**Step 3.** Rights Explanation: LLM synthesizes a jargon-free explanation of the user's statutory rights, obligations of the opposite party, and the remedy available.

**Step 4.** Structured Output: Response is formatted with: Rights Summary | Governing Law | Steps to Enforce | Escalation Path.

**Step 5.** Disclaimer Injection: Mandatory disclaimer appended to every response by the safety middleware.

#### ■ Suggested Tech Stack

Python

LangChain

Pinecone

OpenAI / Claude API

FastAPI

React

#### ■ APIs / Integrations

- Ministry of Labour & Employment — Labour Law portal for wage grievances
- National Consumer Disputes Redressal Commission (NCDRC) — edaakhil.nic.in API
- National Human Rights Commission (NHRC) — complaint status webhook

#### ■ Acceptance Criteria

- System correctly identifies legal theme with  $\geq 90\%$  accuracy on a 200-query benchmark.
- Every response cites at least one specific section of the applicable Indian act.
- Response time  $\leq 4$  seconds for 95th percentile of production traffic.
- Disclaimer present in 100% of rights-explanation responses (automated test).

## ■ Module 2 — Cybercrime Guidance & Complaint Assistance

## M 2

## Cybercrime Guidance Engine

### ■ Scope

- Online Financial Fraud (UPI fraud, phishing, fake investment schemes)
- Identity Theft & Account Takeover
- Cyberstalking & Online Harassment
- Ransomware, Malware & Data Extortion
- Cyberbullying (especially targeting minors)

### ■ ■ Functional Requirements & Actions

**Step 1.** Crime Categorization: Multi-label classifier identifies cybercrime sub-type from user's narrative (supports overlapping categories e.g., 'phishing AND financial fraud').

**Step 2.** Legal Section Mapping: Maps offense to IT Act, 2000 sections (e.g., Sec. 66C — Identity Theft, Sec. 66D — Cheating by personation) and BNS equivalents.

**Step 3.** Complaint Walkthrough: Generates a chronological, numbered guide for filing on cybercrime.gov.in — account creation, report category, evidence upload, acknowledgment.

**Step 4.** Evidence Checklist: Dynamically generates a context-specific evidence list (bank transaction IDs, wallet screenshots, sender metadata, URL with timestamp).

**Step 5.** 1930 Helpline Trigger: If financial fraud detected AND amount > threshold configured, surface the National Cyber Crime Helpline (1930) as the first priority action.

**Step 6.** Follow-Up Tracking: Generate a reference ID checklist template for the user to track their complaint status on the portal.

### ■ Suggested Tech Stack

Python

spaCy NER

LangChain RAG

Redis (session)

FastAPI

Twilio (SMS)

### ■ APIs / Integrations

- National Cyber Crime Reporting Portal — cybercrime.gov.in (deep-link generation)
- RBI DAKSH Portal — for banking fraud escalation
- TRAI DND / SMS alert system for fraud number verification

### ■ Acceptance Criteria

- Cybercrime type correctly classified for  $\geq 88\%$  of test cases across 6 crime categories.
- Complaint walkthrough covers all mandatory steps verified against cybercrime.gov.in live flow.
- Evidence checklist is dynamically tailored — no generic lists (validated by legal SME).
- 1930 helpline surfaces within top 2 elements for any financial fraud query.

## ■ ■ Module 3 — Women's Safety & Legal Support

# M 3

## Women's Safety & Emergency Legal Response

### ■ Scope

- Domestic Violence — Protection of Women from Domestic Violence Act, 2005
- Workplace Sexual Harassment — POSH Act, 2013 (Sexual Harassment of Women at Workplace Act)
- Cyberstalking targeting women — IT Act Sec. 66A / BNS provisions
- Emergency Protection Orders & Shelter access
- NCW Complaint Filing & District Protection Officer access

### ■ Functional Requirements & Actions

**Step 1.** Danger Signal Detection: Sentiment & keyword classifier flags physical-danger indicators ('hitting me', 'afraid to go home', 'threatening') and routes to EMERGENCY MODE.

**Step 2.** Emergency Header: In EMERGENCY MODE, response begins with verified helpline numbers (112, 1091, 181) BEFORE any legal information — non-negotiable UX rule.

**Step 3.** Zero FIR Guidance: Explains BNSS provision for Zero FIR — any police station must register, regardless of jurisdiction — with step-by-step filing instructions.

**Step 4.** Protection Officer Locator: Integrates district Protection Officer directory (Ministry of Women & Child Development data) to surface nearest officer.

**Step 5.** POSH Complaint Flow: For workplace harassment, explains ICC (Internal Complaints Committee) obligation, 90-day filing window, and escalation to LCC if no ICC exists.

**Step 6.** NCW Filing Assistance: Guides user through ncwapps.nic.in online complaint form with required document checklist.

**Step 7.** Safe Language Protocol: Responses avoid re-traumatizing language; use trauma-informed communication patterns reviewed by licensed counsellors.

### ■ Suggested Tech Stack

Python

BERT Sentiment

LangChain

Google Maps API

FastAPI

Flutter

### ■ APIs / Integrations

- Ministry of Women & Child Development — Protection Officer directory API
- NCW Online Complaint Portal — ncwapps.nic.in
- 112 India Emergency App — deep-link for SOS trigger
- iCall / Vandrevalla Foundation — counselling referral API

### ■ Acceptance Criteria

- Danger detection recall  $\geq 95\%$  on annotated test set (zero tolerance for false negatives).
- Emergency helpline numbers appear BEFORE legal content in 100% of flagged responses.
- POSH flow covers all 7 mandatory ICC steps per Act Schedule.

- Zero FIR explanation validated as legally accurate by empanelled lawyer.
- All response text passes trauma-informed language audit with < 2 flagged phrases per 100.

## ■ Module 4 — Tenant-Landlord & Basic Compliance

### M4 Tenant-Landlord Disputes & Business Compliance

#### ■ Scope

- Rental Disputes — Model Tenancy Act 2021 & state-specific Rent Control Acts
- Security Deposit Rules — cap limits, return timelines, deduction rights
- Eviction Procedures — notice periods, grounds, and due process
- Startup Compliance Checklist — GST registration, Trademark, Company incorporation basics
- Consumer rights for homebuyers under RERA (Real Estate Regulation Act)

#### ■ Functional Requirements & Actions

**Step 1.** State Detection: Infer or ask user's state to apply correct Rent Control Act (e.g., Maharashtra Rent Control Act vs Delhi Rent Act vs MTA 2021 defaults).

**Step 2.** Issue Classification: Identify dispute type — deposit withholding, illegal eviction, rent increase dispute, maintenance non-compliance — and surface specific provisions.

**Step 3.** Security Deposit Calculator: Interactive module — input monthly rent → output permissible deposit cap under applicable state law + interest entitlement formula.

**Step 4.** Notice Period Guide: Explain mandatory notice periods for both landlord and tenant across common scenarios (non-payment, lease expiry, sale of property).

**Step 5.** Eviction Rights Explainer: Clarify grounds under which landlord CAN and CANNOT evict; explain self-help eviction prohibition and legal remedy via Rent Authority.

**Step 6.** Business Compliance Checklist: Generate role-specific compliance checklist (Sole Proprietor / LLP / Pvt Ltd) covering GST, PAN, TAN, Trademark, Shops Act.

**Step 7.** RERA Complaint Path: For homebuyer disputes, guide to state RERA portal complaint registration with required documents list.

#### ■ Suggested Tech Stack

Python

LangChain

PostgreSQL (state law DB)

FastAPI

React

Chart.js

#### ■ APIs / Integrations

- State RERA portals (dynamic URL mapping per state)
- MCA21 Portal — company incorporation status verification
- GST Portal — gstin.gov.in for registration status check
- IP India — trademark search API

## ■ Acceptance Criteria

- State-law selection correctly applied for all 29 states + 8 UTs in test matrix.
- Security deposit calculator output matches manual legal calculation for 100 test cases.
- Business compliance checklist covers all mandatory registrations for each entity type.
- RERA complaint path includes correct portal URL for  $\geq 25$  states.

## ■ Module 5 — Legal Document Simplifier

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### Legal Document Simplifier & Risk Analyzer

#### ■ Scope

- Rental Agreements — lease deed analysis, clause-by-clause breakdown
- Employment Contracts — probation terms, notice pay, IP assignment clauses
- Non-Disclosure Agreements (NDAs) — scope, duration, carve-outs, penalties
- Service Agreements & SaaS Terms — liability caps, SLA, data ownership
- Basic Affidavits — format validation and content explanation

## ■ Functional Requirements & Actions

**Step 1.** Document Ingestion: Accept PDF / DOCX / pasted text. Parse and chunk document into logical clause segments using structural NLP (heading detection + boundary rules).

**Step 2.** Clause Classification: Tag each clause by type — Payment, Termination, Penalty, Confidentiality, Jurisdiction, IP Assignment, Indemnity, Limitation of Liability.

**Step 3.** Plain-English Translation: For each clause, generate a  $\leq 3$ -sentence plain summary explaining 'what this means for YOU' in non-legal language.

**Step 4.** Obligation Extraction: Structured output of WHO owes WHAT by WHEN — two columns: 'Your Obligations' and 'Their Obligations'.

**Step 5.** Risk Flag Engine: Automated detection of high-risk patterns — extreme penalty clauses ( $> 5x$  contract value), unilateral termination without notice, perpetual IP assignment, jurisdiction in distant/inconvenient court, auto-renewal without notification.

**Step 6.** Risk Scoring Dashboard: Visual risk summary — Green (standard) / Amber (review) / Red (negotiate or reject) — for each clause category.

**Step 7.** Negotiation Tips: For each Red-flagged clause, suggest a standard market-norm counter-clause language the user can propose.

## ■ Suggested Tech Stack

Python

pdfplumber

spaCy

LangChain

LLM API

React

D3.js

## ■ APIs / Integrations

- No external govt APIs required for core module



- Optional: Bar Council of India — lawyer referral for flagged documents
- DocuSign API — for e-signature after review (Phase 2)

■ Acceptance Criteria

- Clause segmentation correctly identifies ≥ 92% of clause boundaries in 50 test contracts.
- Plain-English summaries rated 'clear' or 'very clear' by ≥ 85% of lay-user test panel.
- Risk flag engine identifies all 6 high-risk pattern types in seeded test documents.
- Obligation extraction produces zero false attributions in 30 dual-party agreement tests.

■■ Cross-Cutting Module — Guardrails & Safety

All 5 feature modules share a centralized Safety & Guardrail Middleware layer. This layer is non-bypassable and runs as a post-processing step on every LLM response before it is returned to the user.

| Guardrail                  | Trigger Condition                                                           | Action                                                                                       |
|----------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Mandatory Disclaimer       | Any response containing legal pathways, actions, or advice                  | Append standardized disclaimer text — 100% coverage enforced by automation                   |
| No Case Outcome Prediction | LLM output contains phrases like 'you will win/lose' or 'guaranteed result' | Flag for review; replace with: 'Legal outcomes are determined by a court or arbitrator'      |
| Emergency Interception     | Danger keywords detected (domestic violence, self-harm, threats)            | Pre-emptively inject crisis helpline (112, 199, 1091, iCall) before all other content        |
| Hallucination Guard        | LLM cites a section not found in the RAG-retrieved sources                  | Flag for review; replace with: 'Please consult a legal professional for this specific point' |
| PII Redaction              | Response accidentally echoes user-provided PII (e.g., name, phone, email)   | Redact, and if applicable, mask sensitive data before delivery                               |

Standard Disclaimer Text (Injected Verbatim)

Disclaimer: LegalEase AI is an AI legal information assistant, not a lawyer. This information is for educational purposes and awareness under Indian law. For formal legal advice, please consult a registered legal professional.

## ■ Non-Functional Requirements

| Category      | Requirement                          | Target                           |
|---------------|--------------------------------------|----------------------------------|
| Performance   | P95 API response time                | ≤ 4 seconds                      |
| Availability  | Uptime SLA                           | 99.5% monthly                    |
| Scalability   | Concurrent users (peak)              | 10,000 CCU                       |
| Accuracy      | Legal citation accuracy (human eval) | ≥ 90%                            |
| Safety        | Emergency interception recall        | ≥ 99% (zero-miss policy)         |
| Privacy       | PII in logs                          | Zero — redacted at gateway       |
| Accessibility | WCAG 2.1 compliance level            | AA                               |
| Multilingual  | Supported Indian languages at launch | Hindi, Malayalam, Tamil, Bengali |
| Audit         | Response logging for legal review    | 100% with 90-day retention       |
| Security      | Penetration test cadence             | Quarterly   OWASP Top 10         |

## ■■ Deployment Roadmap

| Phase                  | Duration    | Deliverables                                                                |
|------------------------|-------------|-----------------------------------------------------------------------------|
| Phase 0 — Foundation   | Weeks 1–2   | Legal corpus ingestion, vector DB setup, RAG baseline, intent classifier v1 |
| Phase 1 — Core Modules | Weeks 3–6   | M1 (Rights), M2 (Cyber), M3 (Women's Safety) — functional + tested          |
| Phase 2 — Extended     | Weeks 7–9   | M4 (Tenant), M5 (Document Simplifier) — full feature + risk engine          |
| Phase 3 — Safety Layer | Week 10     | All guardrails live, PII redaction, emergency interception QA               |
| Phase 4 — Multilingual | Weeks 11–12 | Hindi + Malayalam translation layer, regional legal nuance tuning           |
| Phase 5 — Hardening    | Weeks 13–14 | Load testing, security pentest, legal SME audit, WCAG audit                 |
| Phase 6 — GA Launch    | Week 15     | Production deployment, monitoring dashboards, user feedback loop            |